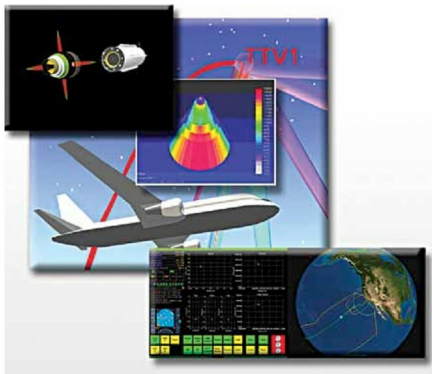


Ensuring Quality and Reliability of Electronic Systems

Learn about the design steps to be followed to make a quality and reliable product



(Image courtesy: www.meicompny.com)

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Key to success of any product lies in its producibility, quality and reliability. Quality and reliability cannot be achieved by just arriving at a schematic to perform a required function and specifying component values. Many more steps are required to make a reliable product. Here we describe the design flow required to achieve a quality and reliable product, followed by other requirements.

Design methodology

Determine. Determine the product requirements such as:

1. Functions that the product has to perform. These functions are called requirements.
2. Operating conditions under which the product will perform its functions. These include temperature range, vibrational stresses and electromagnetic compatibility requirements. The operating condition is called lifecycle environment.
3. Conditions under which the product is to be stored and transported.
4. Power requirements (alternating current or direct current, voltage value, etc).
5. Electrical interface requirements (interfacing signals (analogue/digital),

voltage range for analogue signals, serial input/output or parallel interface standard for digital signals).

6. Mechanical interface requirements (size, shape, material and finish).

7. Regulatory requirements, say, electromagnetic emissions.

Define. Define how best the unit should perform its function in terms of technical performance specifications—power output for a radio transmitter, sensitivity for a radio receiver and non-linearity for a measuring instrument.

Explore design concepts. Explore and evaluate alternative design concepts to meet the requirements determined.

Component selection. Select proven, standard and replaceable components from reputed manufacturers. If components with new technologies or sources are used, evaluate their performance for required functionality under required operational stresses (electrical, mechanical, thermal, etc).

Components should also be able to withstand stresses from handling, storage and assembly. Easy assembly should be possible. Care should be taken so that selected components do not become obsolete very soon. Design should use replaceable modules so that the unit can be repaired and maintained easily.

Performance analysis with electronic design automation (EDA) tools. Simulate the designed schematics with EDA tools. Initially use nominal values for all components in simulation. Confirm that the design is giving required outputs with required inputs. Correct any non-conformance by modifying the design and simulating again.